

MXCuBE Status Report MANACÁ - LNLS (Sirius)

Rafael Lyra, Pedro Benetton
*Beamline Support Group/Staff Members
On Behalf of LNLS (Sirius)*



Current State

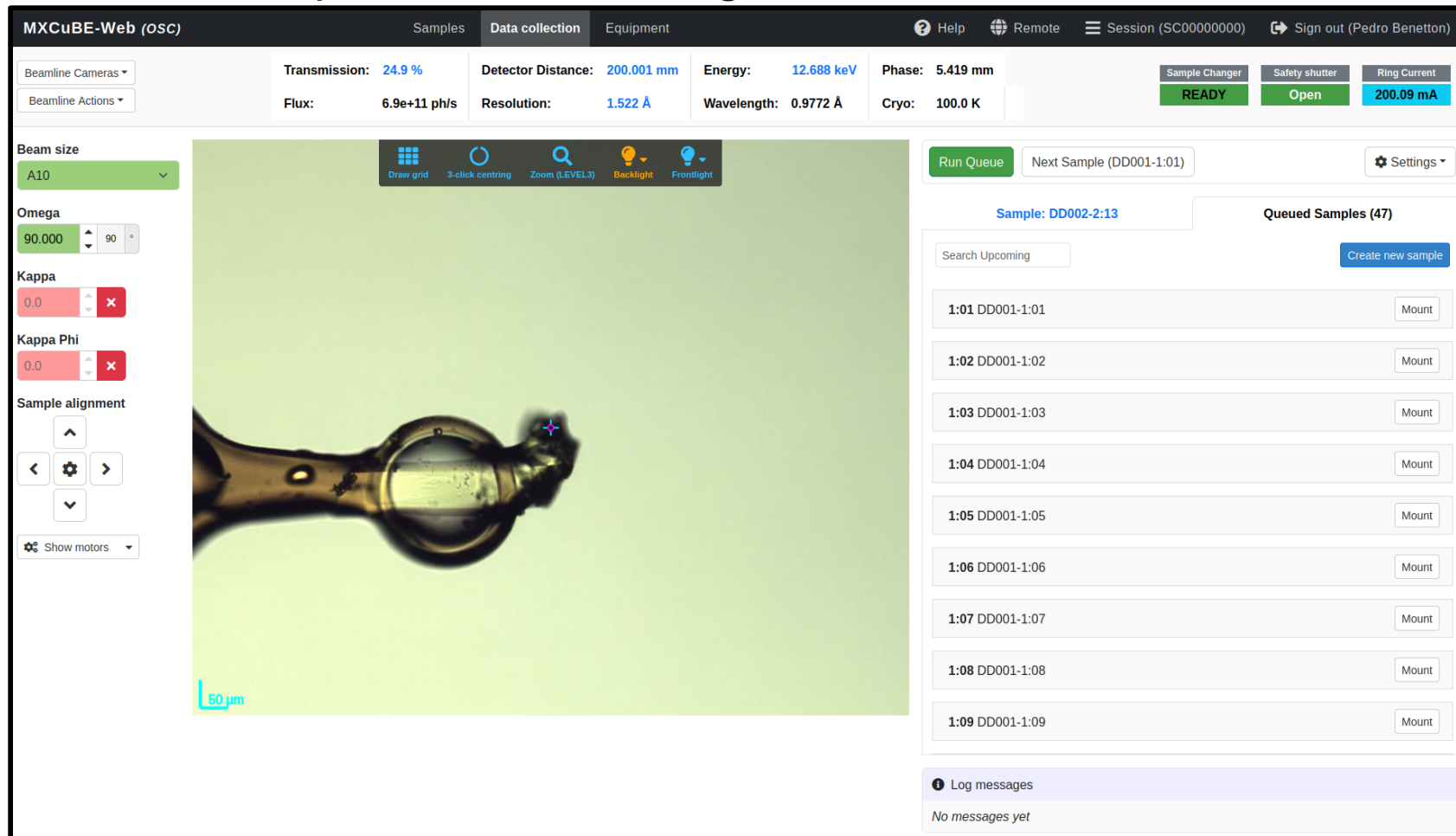
- MXCuBE3 has reached a stable version and will be used only as a backup
- MXCuBE4 -> developed and used at beamline
- New detector! Data collection is now drastically faster (PILATUS4 4M XE)



MXCuBE4 Web

Currently in Production

- Initial Version implemented and being used at beamline



The screenshot displays the MXCuBE-Web (osc) interface, which is used for controlling the MXCuBE4 beamline. The interface is divided into several sections:

- Top Bar:** Includes tabs for "Samples", "Data collection", and "Equipment". It also features a "Help" icon, a "Remote" icon, a session identifier "Session (SC00000000)", and a "Sign out (Pedro Benetton)" button.
- Beamline Status:** A row of buttons for "Beamline Cameras", "Beamline Actions", "Sample Changer", "Safety shutter", and "Ring Current".
- Parameters:** A section displaying key beamline parameters: Transmission: 24.9 %, Flux: 6.9e+11 ph/s, Detector Distance: 200.001 mm, Resolution: 1.522 Å, Energy: 12.688 keV, Wavelength: 0.9772 Å, Phase: 5.419 mm, Cryo: 100.0 K. Status indicators show "READY", "Open", and "200.09 mA".
- Beam size:** A dropdown menu set to "A10".
- Omega:** A numeric input field set to 90.000, with a "90" button.
- Kappa:** A numeric input field set to 0.0, with a red "X" button.
- Kappa Phi:** A numeric input field set to 0.0, with a red "X" button.
- Sample alignment:** A section with directional arrows and a "Show motors" checkbox.
- Central Camera View:** A large image showing a sample mounted on a goniometer. A scale bar indicates 50 µm.
- Right Panel:** Contains a "Run Queue" button, a "Next Sample (DD001-1:01)" button, and a "Settings" dropdown. Below this is a "Sample: DD002-2:13" section and a "Queued Samples (47)" list. The list includes sample IDs and "Mount" buttons for each.
- Log messages:** A section at the bottom right with a "Log messages" button and a status "No messages yet".

MXCuBE4 Web

Login and Proposal Selection



CNPq

- Integrated with our local instance of ICAT



Select a session

Filter

Active (5) Scheduled (9) Other beamlines (0)

ID	Title	Start	End	Portal	User	Logbook
RAP-20254581	Protein-Ligand Interactions of Inflammation-Related Enzymes PKM2 and PAD4	14-05-2026 11:01:00	15-05-2026 11:00:00	🔗	🔗	🔗
RAP-20253570	Structural studies of carbohydrate-active enzymes	13-05-2026 11:01:00	14-05-2026 11:00:00	🔗	🔗	🔗
Industrial-20232083	Desenvolvimento de analgésico para o tratamento de dor neuropática a partir da biodiversidade brasileira	01-05-2026 03:01:00	16-05-2026 03:00:00	🔗	🔗	🔗
SC-00000000	Scientific Commissioning of the MANACÁ Beamline.	01-01-2026 08:00:00	01-01-2029 18:00:00	🔗	🔗	🔗
TEST-0020	Investigation under embargo till 2050.	05-10-2025 11:00:00	06-12-2026 11:00:00	🔗	🔗	🔗

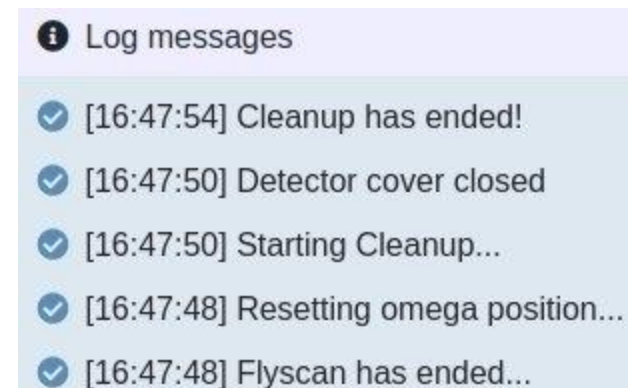
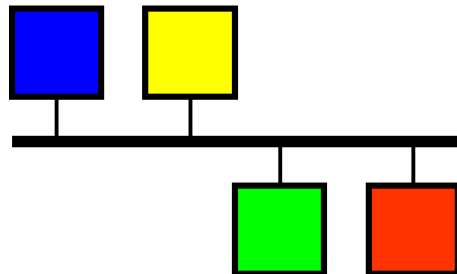
Select SC-00000000 Cancel

MXCuBE4 Web

Beamline Orchestration

- EPICS Abstract Classes: EPICSMotor, EPICSActuator, EPICSNState
- Other specific classes inherit from these for specific cases, when needed
- EPICS orchestration with Bluesky (HTTP)
- Bluesky logs are streamed to MXCuBE logs
- MXCuBE Queue Manager is connected with Bluesky Queue Server

EPICS



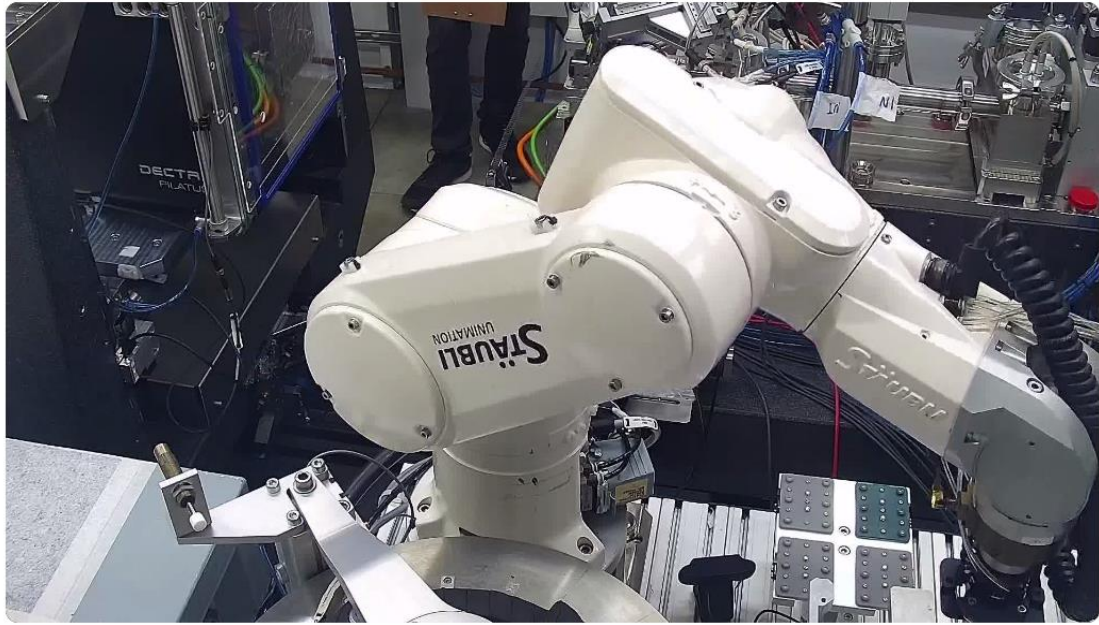
MXCuBE4 Web

Beamline Cameras

- Accessible directly from MXCuBE

**Manacá Beamline**
Hutch Surveillance Camera

Cabana B ● Cabana A



⚙️ Controls

Directions

↶ ↷ ↵

⏪ ⏩ ⏴

⏴ ⏵ ⏶

Zoom

⏪ ⏩

Speed

⏪ 0.2 ⏩

⏴ Set Home

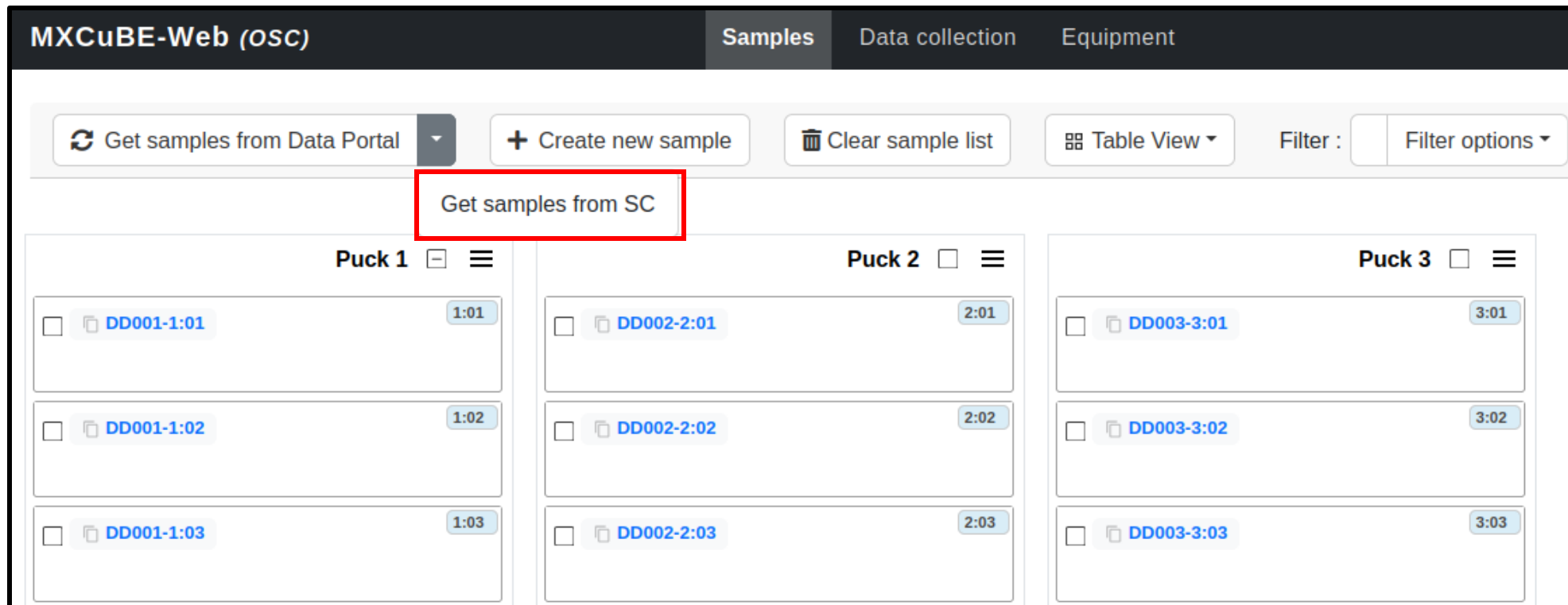
Webcams ⌵ ✕



MXCuBE4 Web

Sample Changer

- Using "Get samples from SC" option while sample register via ICAT is not implemented



MXCuBE-Web (OSC)

Samples Data collection Equipment

Get samples from Data Portal Create new sample Clear sample list Table View Filter : Filter options

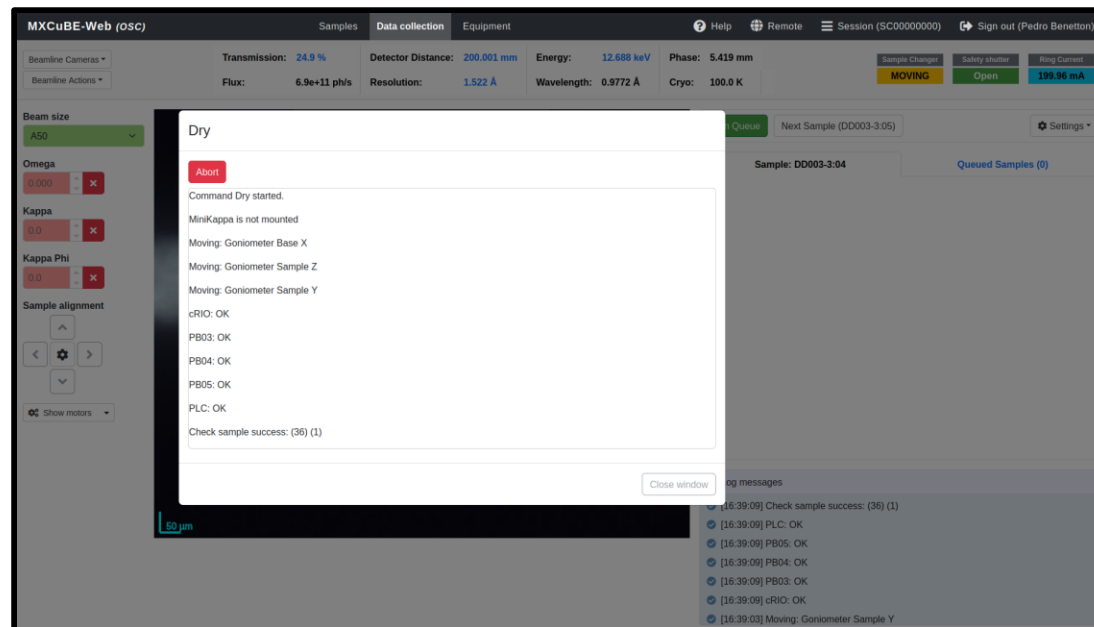
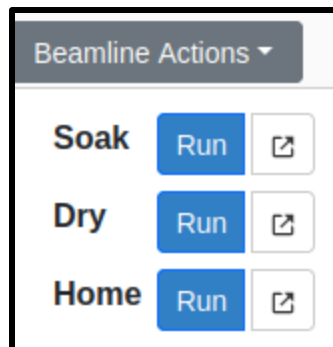
Get samples from SC

Puck 1	Puck 2	Puck 3
☐ DD001-1:01 1:01	☐ DD002-2:01 2:01	☐ DD003-3:01 3:01
☐ DD001-1:02 1:02	☐ DD002-2:02 2:02	☐ DD003-3:02 3:02
☐ DD001-1:03 1:03	☐ DD002-2:03 2:03	☐ DD003-3:03 3:03

MXCuBE4 Web

Sample Changer

- Soak, Dry and Home are callable at Beamline Actions menu
- Mount and Unmount are also defined at the Beamline Actions class
- All Beamline Actions are called via Bluesky Plans
- UI blocks all input while robot is moving



Data Collection

- Flyscan, Energy Scan, XRF, Grid Scan (All via Bluesky)

XRF

Path:

/ibira/Inls/beamlines/manaca/proposals/00000000/data/20260515/DD006-3-08/

Filename:

Subdirectory

DD006-3-08/

Prefix

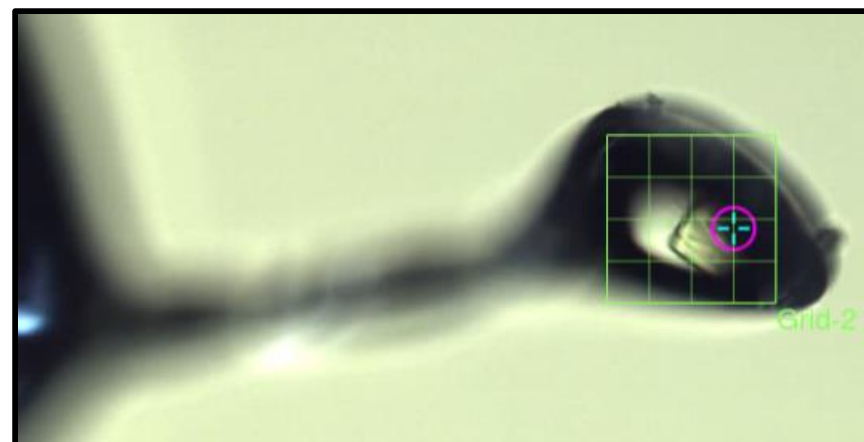
DD006-3-08

Count time (s)

0.01

Run Now

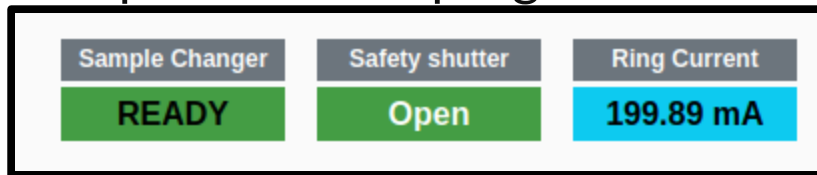
Add to Queue



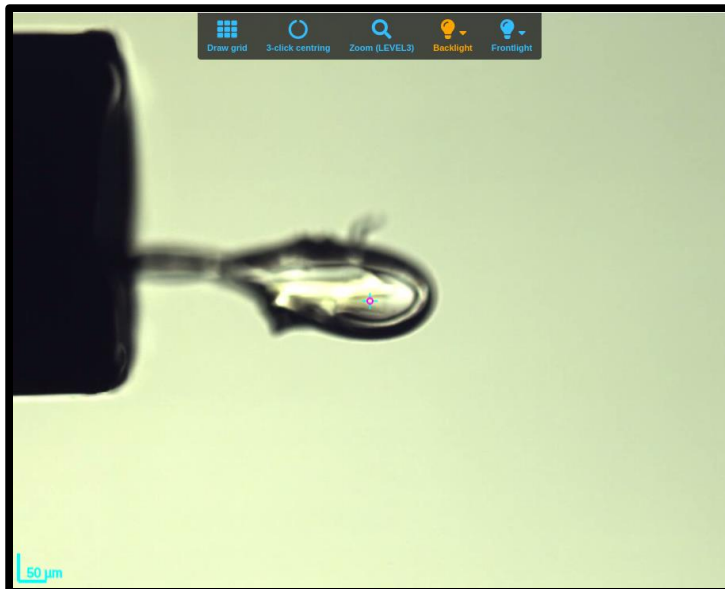
MXCuBE4 Web

Other Features

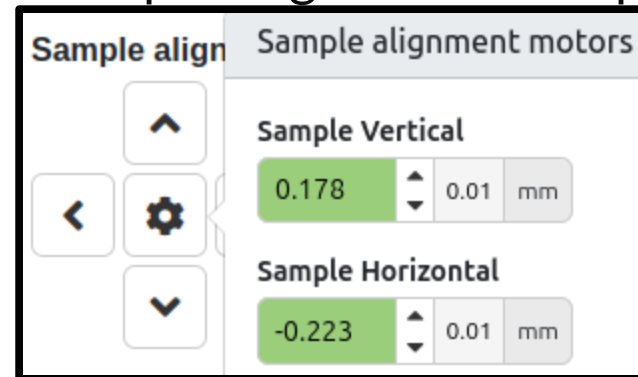
- Components at top-right corner



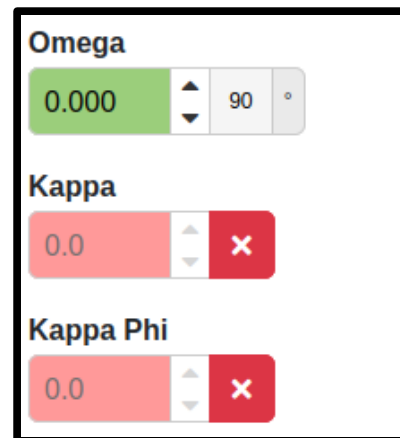
- Video Streamer with EPICS PVA



- Sample Alignment Tooltip



- EPICSMotorDetachable



MXCuBE4 Web

Code Changes



CNPq

- Using AbstractDiffractometer and SampleView
- Drastic amount of code reduced in the transition from version 3 to 4, the main reasons being the EPICS classes and code cleanup
- MXCuBE4 main branch is ahead of official repo's develop branch
- We plan to rebase every month, or right before starting a new PR
- Subject-focused, small PRs

LIMS

- Application that allows quick access to data processing results

Web Puck ID Puck ID new Help

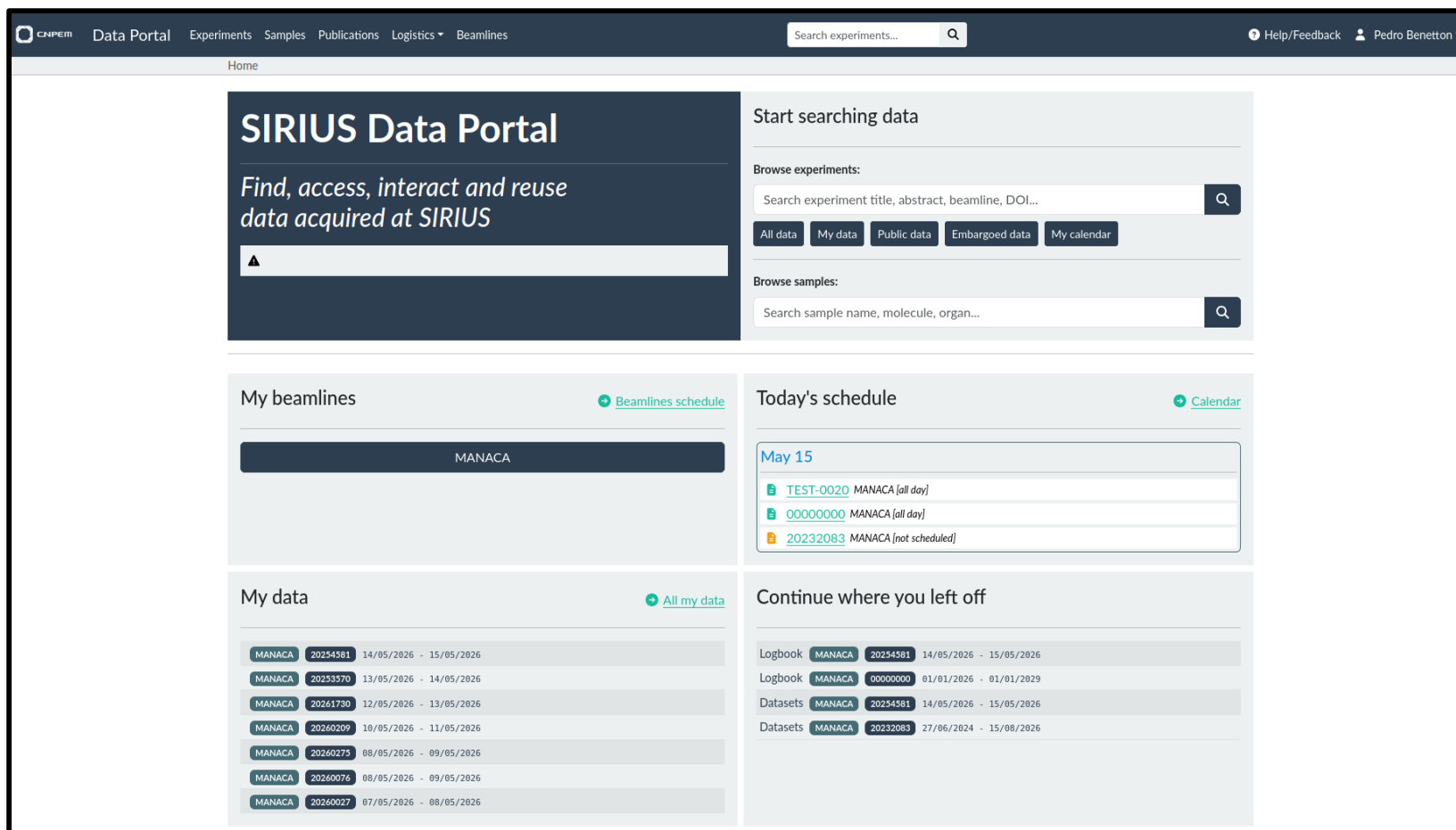
20232083.xlsx Data Modal

nA) fI	Folder Name fI	Loop image fI	Processing Type fI	Collection Type fI	SG fI	a fI	b fI	c fI	alpha fI	beta fI	gama fI	Resolution fI	I/SIGMA fI
	DD001-1-12		fragment_screening	Data Collection	4	69.47	51.45	135.37	90.000	103.099	90.000	2.25	1.33
	DD001-1-13		fragment_screening	Data Collection	4	69.96	51.92	136.12	90.000	103.299	90.000	2.34	1.40
	DD001-1-14		fragment_screening	Data Collection	4	69.82	51.69	135.62	90.000	103.247	90.000	2.45	1.26
	DD001-1-15		fragment_screening	Data Collection	4	69.82	51.82	135.85	90.000	103.349	90.000	2.30	1.44
	DD001-1-16		fragment_screening	Data Collection	4	69.98	52.09	136.06	90.000	103.459	90.000	2.25	1.75
	DD002-2-01		fragment_screening	Data Collection	3	70.04	51.63	135.95	90.000	103.101	90.000	2.22	1.39
	DD002-2-02		fragment_screening	Data Collection	4	70.14	51.74	136.28	90.000	103.181	90.000	2.09	1.40
	DD002-2-03		fragment_screening	Data Collection	4	70.28	51.83	136.53	90.000	103.237	90.000	2.04	1.41
	DD002-2-04		fragment_screening	Data Collection	4	70.01	51.62	135.88	90.000	103.268	90.000	2.55	1.40
	DD002-2-05		fragment_screening	Data Collection	4	69.73	51.67	135.91	90.000	103.146	90.000	2.09	1.41
	DD002-2-06		fragment_screening	Data Collection	4	69.85	51.86	135.88	90.000	103.372	90.000	1.89	1.39
	DD002-2-07		fragment_screening	Data Collection	4	70.28	51.78	136.25	90.000	103.158	90.000	2.25	1.34
	DD002-2-08		fragment_screening	Data Collection	4	69.90	51.63	135.71	90.000	103.058	90.000	2.22	1.52
	DD002-2-09		fragment_screening	Data Collection	4	69.88	51.59	135.67	90.000	103.042	90.000	2.19	1.45
	DD002-2-10		fragment_screening	Data Collection	4	69.93	51.62	135.63	90.000	103.195	90.000	2.36	1.39
	DD002-2-11		fragment_screening	Data Collection	3	69.75	51.51	135.58	90.000	103.264	90.000	4.08	1.18

Filter: Filter by day: 15/05/2026 Download: Refetch:

LIMS

- Local Instance of ICAT developed by data science management group (allan.pinto@lnls.br)



The screenshot displays the SIRIUS Data Portal interface. The top navigation bar includes links for Data Portal, Experiments, Samples, Publications, Logistics, and Beamlines, along with a search bar and user information (Pedro Benetton). The main content area is divided into several sections:

- SIRIUS Data Portal**: A large header section with the tagline "Find, access, interact and reuse data acquired at SIRIUS".
- Start searching data**: A section with search filters for experiments and samples, including buttons for "All data", "My data", "Public data", "Embargoed data", and "My calendar".
- My beamlines**: A section showing a list of beamlines, currently displaying "MANACA".
- Today's schedule**: A section showing the schedule for May 15, listing experiments like "TEST-0020", "00000000", and "20232083".
- My data**: A section showing a list of data entries, including "MANACA" and "20232083".
- Continue where you left off**: A section showing a list of logbooks and datasets, including "MANACA" and "20232083".

Conclusions

Next Steps

- ICAT: integrate with logbook and sample register/ingestion
- Helical Scan
- X-ray centring
- Improve containerization/deployment (podman)
- EPICS Classes: Once we are confident that they are working properly, make them generic in the mxcore repository rather than just in the LNLS folder

Conclusions

Discussion



CNPq

- Challenges: The usage of bluesky seems to be taking some functionalities of MXCuBE, which is now working more as a GUI layer rather than the main actuator on beamline components. Some objects/functions had to be created at mxcube-core just to avoid it from crashing (Adapters, Abstract classes)
- Challenges: properly sending heatmap info from mxcube-core to mxcube-web after gridscan data processing
- Challenges: properly administrating sessions and proposal selection
- PR Control on how mxcube-web gets updated
- Video-streamer EPICS PVA PR

Thank You

Pedro Benetton

pedro.benetton@lnls.br

+55 (19) 99736-1104

MANACÁ beamline

Andrey Nascimento

Guilherme Aun

Control Software

Ana Clara de Souza Oliveira

Rafael Lyra

Data Science and Management

Allan da Silva Pinto

Paulo Baraldi Mausbach

Liu Yi Ling

cnpem.br



CNPem
Brazilian Center for Research
in Energy and Materials

MINISTRY OF
SCIENCE TECHNOLOGY
AND INNOVATION

